

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Receiver
Manufacturer: Arcam
Firmware Version: N/A
Model(s): AVR850, AVR390, AVR550, AV860, SR250

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_0_0	1/17/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

AudioMode	'Analogue'	'Digital'	'HDMI'
['Zone']	'1' – '2'		
HDMIOutput	'1' – '2'	'Both'	
Input	'CD' 'SAT' 'AUX' 'Net' 'Game'	'BD' 'PVR' 'Display' 'USB'	'AV' 'VCR' 'Tuner (FM)' 'STB'
['Zone']	'1' – '2'		
Keypad	'0' – '9'		
['Zone']	'1' – '2'		
MenuNavigation	'Disc (Record) (Enter Trim Menu)' 'Down' 'OK' 'Green'	'System Menu' 'Left' 'Home' 'Yellow'	'Up' 'Right' 'Red' 'Blue'
['Zone']	'1' – '2'		
Mute	'On'	'Off'	
['Zone']	'1' – '2'		
Power	'On'	'Off'	
['Zone']	'1' – '2'		
Transport	'Rewind' 'Next' 'Pause'	'Fast Forward' 'Stop' 'Random'	'Previous' 'Play' 'Repeat'
['Zone']	'1' – '2'		
TuneFM	'Increment'	'Decrement'	
['Zone']	'1' – '2'		
Volume	0 to 99 in steps of 1 dB		
['Zone']	'1' – '2'		
Zone1OnScreenDisplay	'On'	'Off'	

Status Available:

AudioMode	'Analogue'	'Digital'	'HDMI'
['Zone']	'1' – '2'		
ConnectionStatus	'Connected'	'Disconnected'	
HDMIOutput	'1' – '2'	'Both'	
Input	'Follow Zone 1'	'CD'	'BD'
	'AV'	'SAT'	'PVR'
	'VCR'	'AUX'	'Display'
	'Tuner (FM)'	'Net'	'USB'
	'STB'	'Game'	
['Zone']	'1' – '2'		
Mute	'On'	'Off'	
['Zone']	'1' – '2'		
Power	'On'	'Off'	
['Zone']	'1' – '2'		
TuneFMStatus	87 – 108		
['Zone']	'1' – '2'		
Volume	0 to 99 in steps of 1 dB		
['Zone']	'1' – '2'		
Zone1OnScreenDisplay	'On'	'Off'	

Cable and Adapter Requirements:

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device:

Serial communication:

Port Type: RS-232

Baud Rate: 38400

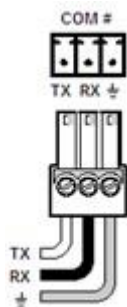
Data Bits: 8

Parity: None

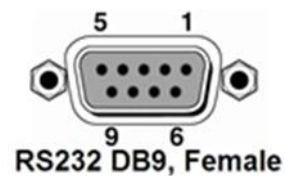
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		3	TxD
RxD		2	RxD
GND		5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	50000
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Audio Mode Analogue Zone 1	!\x01\x0B\x01\x00\x0D
Audio Mode Analogue Zone 2	!\x02\x0B\x01\x00\x0D
Audio Mode Digital Zone 1	!\x01\x0B\x01\x01\x0D
Audio Mode Digital Zone 2	!\x02\x0B\x01\x01\x0D
Audio Mode HDMI Zone 1	!\x01\x0B\x01\x02\x0D
Audio Mode HDMI Zone 2	!\x02\x0B\x01\x02\x0D
HDMI Output 1	!\x01O\x01\x02\x0D
HDMI Output 2	!\x01O\x01\x03\x0D
HDMI Output Both	!\x01O\x01\x04\x0D
Input AUX Zone 1	!\x01\x08\x02\x10c\x0D
Input AUX Zone 2	!\x02\x08\x02\x17\x0D\x0D
Input AV Zone 1	!\x01\x08\x02\x10^\x0D
Input AV Zone 2	!\x02\x08\x02\x17\x09\x0D
Input BD Zone 1	!\x01\x08\x02\x10b\x0D
Input BD Zone 2	!\x02\x08\x02\x17\x07\x0D
Input CD Zone 1	!\x01\x08\x02\x10v\x0D
Input CD Zone 2	!\x02\x08\x02\x17\x06\x0D
Input Display Zone 1	!\x01\x08\x02\x10:\x0D
Input Game Zone 1	!\x01\x08\x02\x10a\x0D
Input Game Zone 2	!\x02\x08\x02\x17\x0B\x0D
Input Net Zone 1	!\x01\x08\x02\x10\x0D
Input Net Zone 2	!\x02\x08\x02\x17\x13\x0D
Input PVR Zone 1	!\x01\x08\x02\x10`\x0D
Input PVR Zone 2	!\x02\x08\x02\x17\x0F\x0D
Input SAT Zone 1	!\x01\x08\x02\x10x1B\x0D
Input SAT Zone 2	!\x02\x08\x02\x17\x14\x0D
Input STB Zone 1	!\x01\x08\x02\x10d\x0D
Input STB Zone 2	!\x02\x08\x02\x17\x08\x0D
Input Tuner (FM) Zone 1	!\x01\x08\x02\x10x1C\x0D
Input Tuner (FM) Zone 2	!\x02\x08\x02\x17\x0E\x0D
Input USB Zone 1	!\x01\x08\x02\x10j\x0D
Input USB Zone 2	!\x02\x08\x02\x17\x12\x0D
Input VCR Zone 1	!\x01\x08\x02\x10w\x0D
Input VCR Zone 2	!\x02\x08\x02\x17\x15\x0D
Keypad 0 Zone 1	!\x01\x08\x02\x10\x00\x0D
Keypad 0 Zone 2	!\x02\x08\x02\x10\x00\x0D
Keypad 1 Zone 1	!\x01\x08\x02\x10\x01\x0D
Keypad 1 Zone 2	!\x02\x08\x02\x10\x01\x0D
Keypad 2 Zone 1	!\x01\x08\x02\x10\x02\x0D
Keypad 2 Zone 2	!\x02\x08\x02\x10\x02\x0D
Keypad 3 Zone 1	!\x01\x08\x02\x10\x03\x0D
Keypad 3 Zone 2	!\x02\x08\x02\x10\x03\x0D
Keypad 4 Zone 1	!\x01\x08\x02\x10\x04\x0D
Keypad 4 Zone 2	!\x02\x08\x02\x10\x04\x0D
Keypad 5 Zone 1	!\x01\x08\x02\x10\x05\x0D

Global Scripter Module Communication Sheet

Revision: 1/18/2017

Keypad 5 Zone 2	!\x02\x08\x02\x10\x05\x0D
Keypad 6 Zone 1	!\x01\x08\x02\x10\x06\x0D
Keypad 6 Zone 2	!\x02\x08\x02\x10\x06\x0D
Keypad 7 Zone 1	!\x01\x08\x02\x10\x07\x0D
Keypad 7 Zone 2	!\x02\x08\x02\x10\x07\x0D
Keypad 8 Zone 1	!\x01\x08\x02\x10\x08\x0D
Keypad 8 Zone 2	!\x02\x08\x02\x10\x08\x0D
Keypad 9 Zone 1	!\x01\x08\x02\x10\x09\x0D
Keypad 9 Zone 2	!\x02\x08\x02\x10\x09\x0D
Menu Navigation Blue Zone 1	!\x01\x08\x02\x107\x0D
Menu Navigation Blue Zone 2	!\x02\x08\x02\x107\x0D
Menu Navigation Disc (Record) (Enter Trim Menu) Zone 1	!\x01\x08\x02\x10Z\x0D
Menu Navigation Disc (Record) (Enter Trim Menu) Zone 2	!\x02\x08\x02\x10Z\x0D
Menu Navigation Down Zone 1	!\x01\x08\x02\x10U\x0D
Menu Navigation Down Zone 2	!\x02\x08\x02\x10U\x0D
Menu Navigation Green Zone 1	!\x01\x08\x02\x10*\x0D
Menu Navigation Green Zone 2	!\x02\x08\x02\x10*\x0D
Menu Navigation Home Zone 1	!\x01\x08\x02\x10+\x0D
Menu Navigation Home Zone 2	!\x02\x08\x02\x10+\x0D
Menu Navigation Left Zone 1	!\x01\x08\x02\x10Q\x0D
Menu Navigation Left Zone 2	!\x02\x08\x02\x10Q\x0D
Menu Navigation OK Zone 1	!\x01\x08\x02\x10W\x0D
Menu Navigation OK Zone 2	!\x02\x08\x02\x10W\x0D
Menu Navigation Red Zone 1	!\x01\x08\x02\x10)\x0D
Menu Navigation Red Zone 2	!\x02\x08\x02\x10)\x0D
Menu Navigation Right Zone 1	!\x01\x08\x02\x10P\x0D
Menu Navigation Right Zone 2	!\x02\x08\x02\x10P\x0D
Menu Navigation System Menu Zone 1	!\x01\x08\x02\x10R\x0D
Menu Navigation System Menu Zone 2	!\x02\x08\x02\x10R\x0D
Menu Navigation Up Zone 1	!\x01\x08\x02\x10V\x0D
Menu Navigation Up Zone 2	!\x02\x08\x02\x10V\x0D
Menu Navigation Yellow Zone 1	!\x01\x08\x02\x10+\x0D
Menu Navigation Yellow Zone 2	!\x02\x08\x02\x10+\x0D
Mute Off Zone 1	!\x01\x08\x02\x10x\x0D
Mute Off Zone 2	!\x02\x08\x02\x17\x05\x0D
Mute On Zone 1	!\x01\x08\x02\x10\x1A\x0D
Mute On Zone 2	!\x02\x08\x02\x17\x04\x0D
Power Off Zone 1	!\x01\x08\x02\x10 \x0D
Power Off Zone 2	!\x02\x08\x02\x17 \x0D
Power On Zone 1	!\x01\x08\x02\x10{\x0D
Power On Zone 2	!\x02\x08\x02\x17{\x0D
Transport Fast Forward Zone 1	!\x01\x08\x02\x104\x0D
Transport Fast Forward Zone 2	!\x02\x08\x02\x104\x0D
Transport Next Zone 1	!\x01\x08\x02\x10\x0B\x0D
Transport Next Zone 2	!\x02\x08\x02\x10\x0B\x0D
Transport Pause Zone 1	!\x01\x08\x02\x100\x0D
Transport Pause Zone 2	!\x02\x08\x02\x100\x0D
Transport Play Zone 1	!\x01\x08\x02\x105\x0D

Transport Play Zone 2	!\x02\x08\x02\x105\x0D
Transport Previous Zone 1	!\x01\x08\x02\x10!\x0D
Transport Previous Zone 2	!\x02\x08\x02\x10!\x0D
Transport Random Zone 1	!\x01\x08\x02\x10L\x0D
Transport Random Zone 2	!\x02\x08\x02\x10L\x0D
Transport Repeat Zone 1	!\x01\x08\x02\x101\x0D
Transport Repeat Zone 2	!\x02\x08\x02\x101\x0D
Transport Rewind Zone 1	!\x01\x08\x02\x10y\x0D
Transport Rewind Zone 2	!\x02\x08\x02\x10y\x0D
Transport Stop Zone 1	!\x01\x08\x02\x106\x0D
Transport Stop Zone 2	!\x02\x08\x02\x106\x0D
Tune FM Decrement Zone 1	!\x01\x16\x01\x00\x0D
Tune FM Decrement Zone 2	!\x02\x16\x01\x00\x0D
Tune FM Increment Zone 1	!\x01\x16\x01\x01\x0D
Tune FM Increment Zone 2	!\x02\x16\x01\x01\x0D
Volume 0 Zone 1	!\x01\x0D\x01\x00\x0D
Volume 0 Zone 2	!\x02\x0D\x01\x00\x0D
Volume 99 Zone 1	!\x01\x0D\x01c\x0D
Volume 99 Zone 2	!\x02\x0D\x01c\x0D
Zone 1 On Screen Display Off	!\x01N\x01\xF2\x0D
Zone 1 On Screen Display On	!\x01N\x01\xF1\x0D

Appendix B. Query Commands

Audio Mode Zone 1	!\x01\x0B\x01\xF0\x0D
Audio Mode Zone 2	!\x02\x0B\x01\xF0\x0D
HDMI Output	!\x01O\x01\xF0\x0D
Heartbeat	!\x01%\x01\xF0\x0D
Input Zone 1	!\x01\x1D\x01\xF0\x0D
Input Zone 2	!\x02\x1D\x01\xF0\x0D
Mute Zone 1	!\x01\x0E\x01\xF0\x0D
Mute Zone 2	!\x02\x0E\x01\xF0\x0D
Power Zone 1	!\x01\x00\x01\xF0\x0D
Power Zone 2	!\x02\x00\x01\xF0\x0D
Tune FM Status Zone 1	!\x01\x16\x01\xF0\x0D
Tune FM Status Zone 2	!\x02\x16\x01\xF0\x0D
Volume Zone 1	!\x01\x0D\x01\xF0\x0D
Volume Zone 2	!\x02\x0D\x01\xF0\x0D
Zone 1 On Screen Display	!\x01N\x01\xF0\x0D